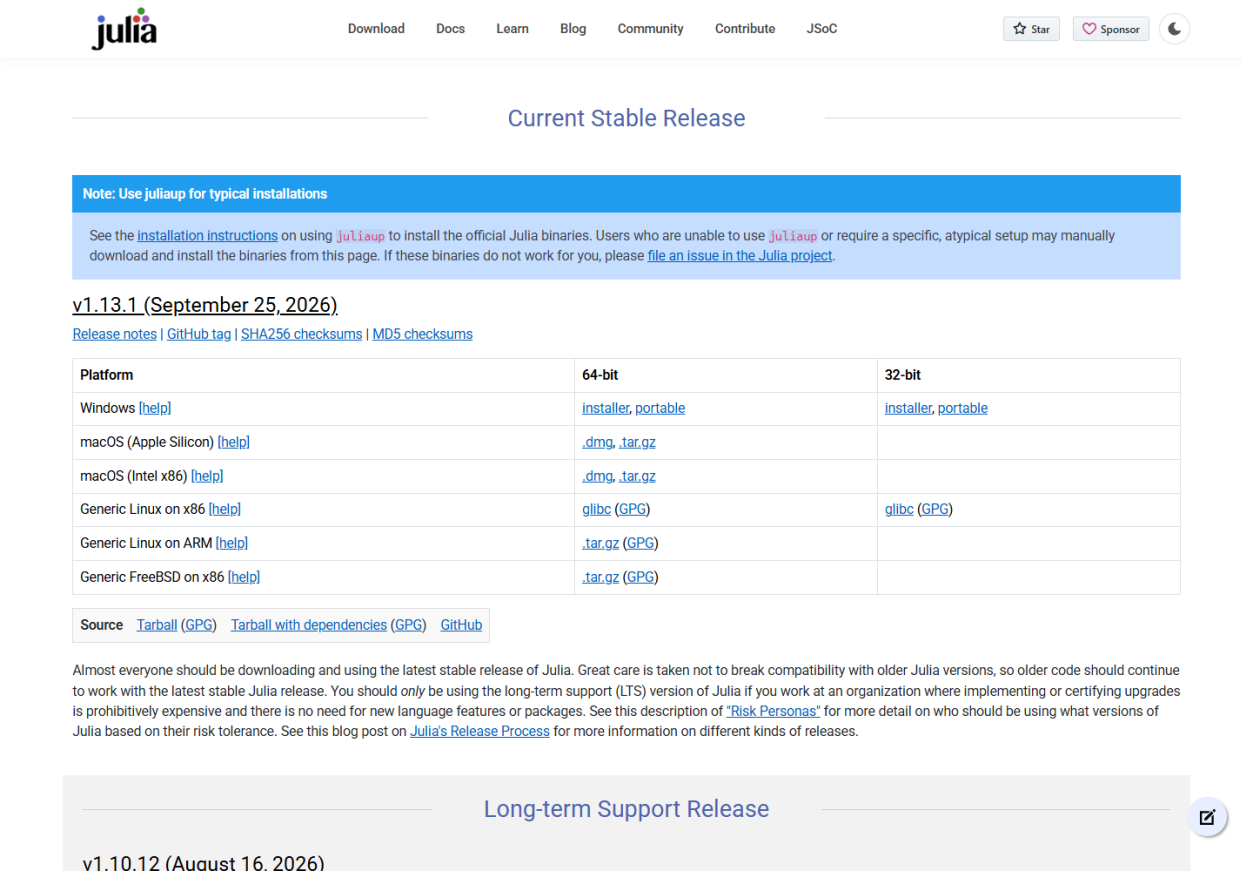
1 Install Julia

Tick Add Julia to PATH. The launchers need it to find Julia.

1. Open julialang.org/downloads/manual-downloads.
2. Download the Windows 64-bit installer for **Julia 1.13.1**, the release tested with this checkout.
3. Run the installer and **tick Add Julia to PATH**. Do not skip this option.
4. Finish installation. Reopen any windows that were already open before installing Julia.

If Julia is already installed without PATH, rerun/modify its installer or follow the official Windows installation instructions.

Ready when: the Julia installation is complete and its executable is on PATH.



Choose the installer for your operating system. The webpage appearance may change.

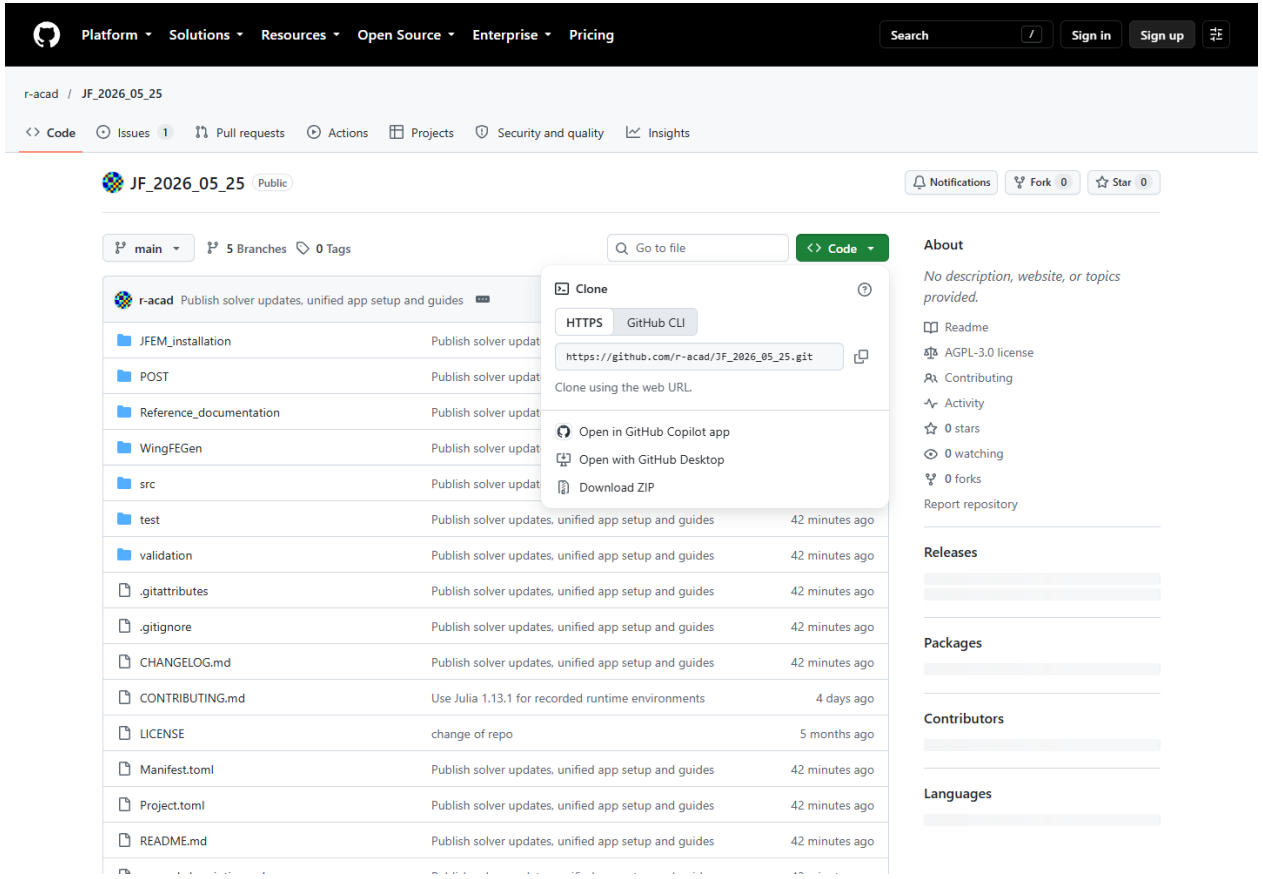
2 Download and extract JFEM

Keep the entire repository together. Do not run it inside the ZIP.

1. Open github.com/r-acad/JF_2026_05_25.
2. Click **Code**, then **Download ZIP**.
3. Right-click the download and choose **Extract All**.
4. Open the extracted folder until you see **setup.cmd**, **JFEM-Boxer.cmd**, **APPS** and `__Starthere!__`.

Choose a writable location. App folders use the shared solver, packages and assets beside them.

Ready when: you are looking at ordinary extracted files, not the ZIP contents.



Download the complete repository using Code, then Download ZIP.

3 Double-click setup.cmd

One setup installs the packages and builds one shared sysimage automatically.

1. Double-click **setup.cmd** in the top folder. No Julia command is needed.

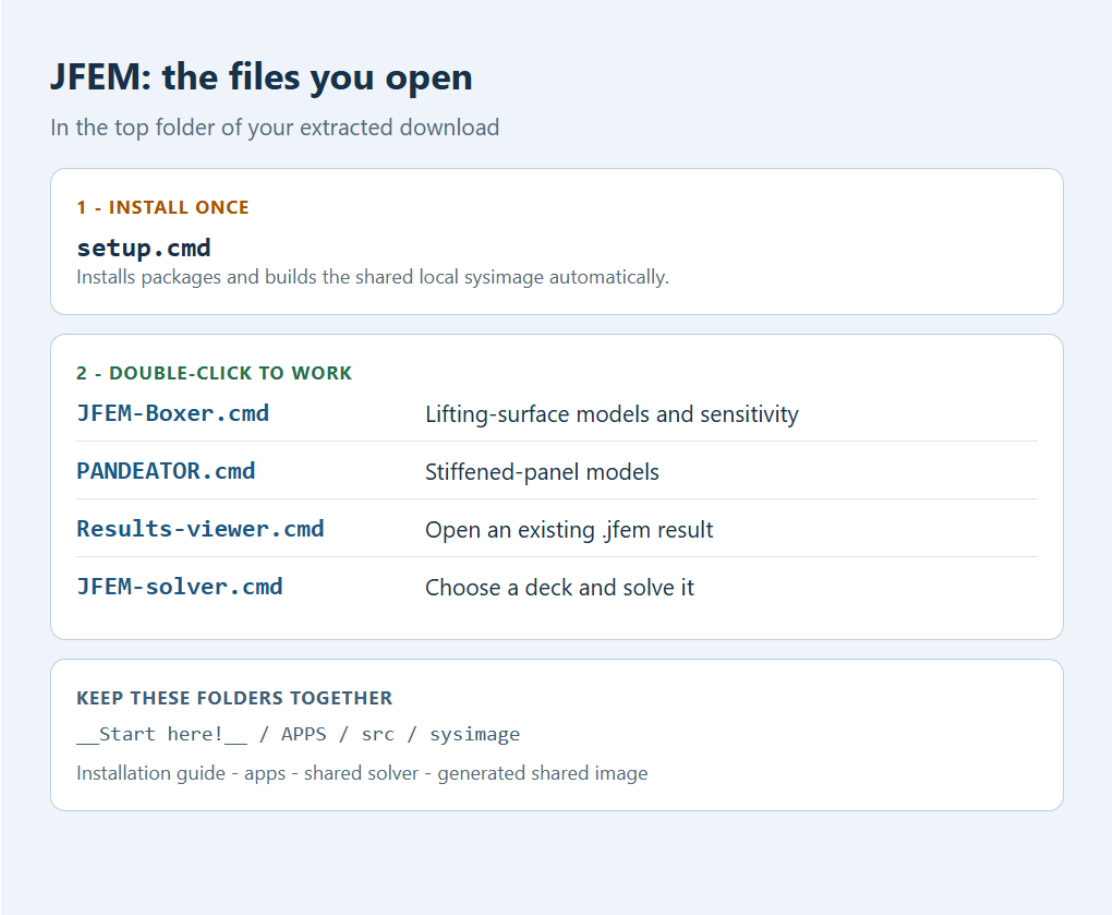
2. Stay connected to the internet. Setup installs packages, checks the apps and precompiles representative workloads.

3. Wait while it builds the **shared JFEM/app sysimage**. The solver and all apps use the same root Project/Manifest and package versions. The first build can take an hour or longer, with long quiet periods during code generation.

4. Keep the window open until setup reports completion. If an error is reported, retain its text and rerun setup after addressing it.

Repeated setup reuses a valid image. No manual package list is required.

Ready when: setup has finished successfully, including the shared local image.



Installation and launch files in the extracted repository.

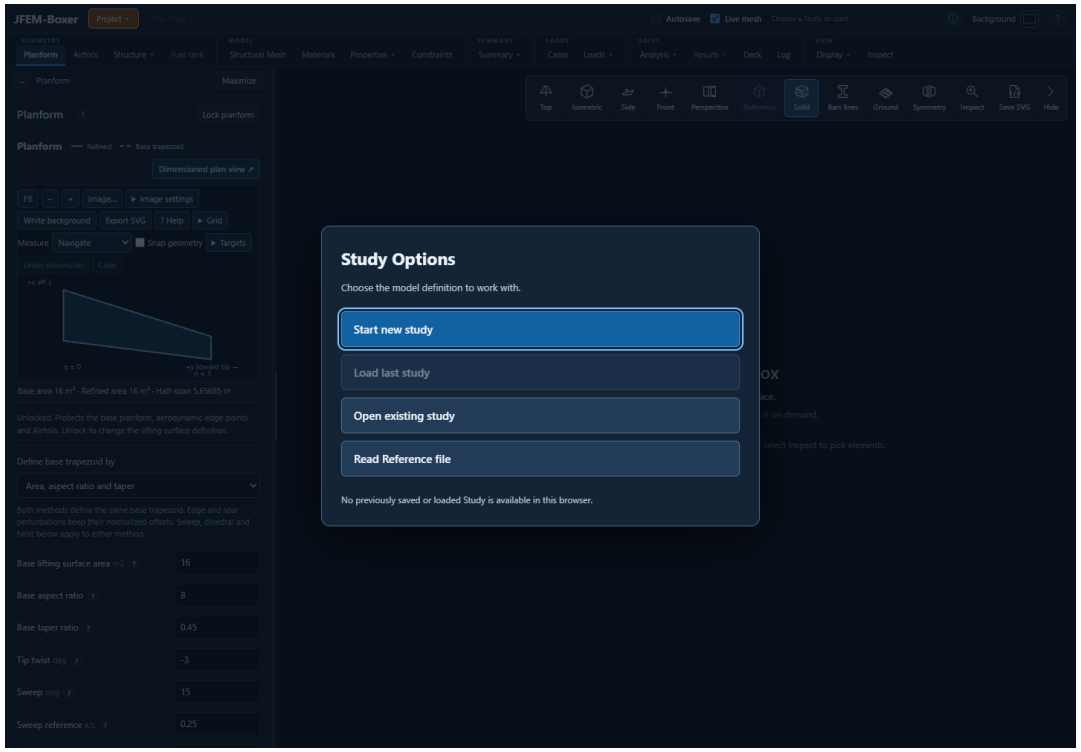
4 Double-click your application

All Windows launchers are in the top folder.

File	Purpose
JFEM-Boxer.cmd	Lifting-surface modeling, loads, FE analysis, sensitivity and results.
PANDEATOR.cmd	Stiffened-panel modeling and analysis.
Results-viewer.cmd	Open existing .jfem results; no Julia server.
JFEM-solver.cmd	Choose an input deck and run JFEM directly; drag-and-drop also works.

In JFEM-Boxer choose **Start new study**, or use **Project** to reopen a saved study or import a Reference deck. Keep the JFEM-Boxer/PANDEATOR console open while working: it runs the local server. The browser address is printed there.

Ready when: the chosen app is open.



JFEM-Boxer starts with Study Options. Choose the appropriate model source.

5 Run and inspect a model

The case, field, deformation scale and legend explain what is being displayed.

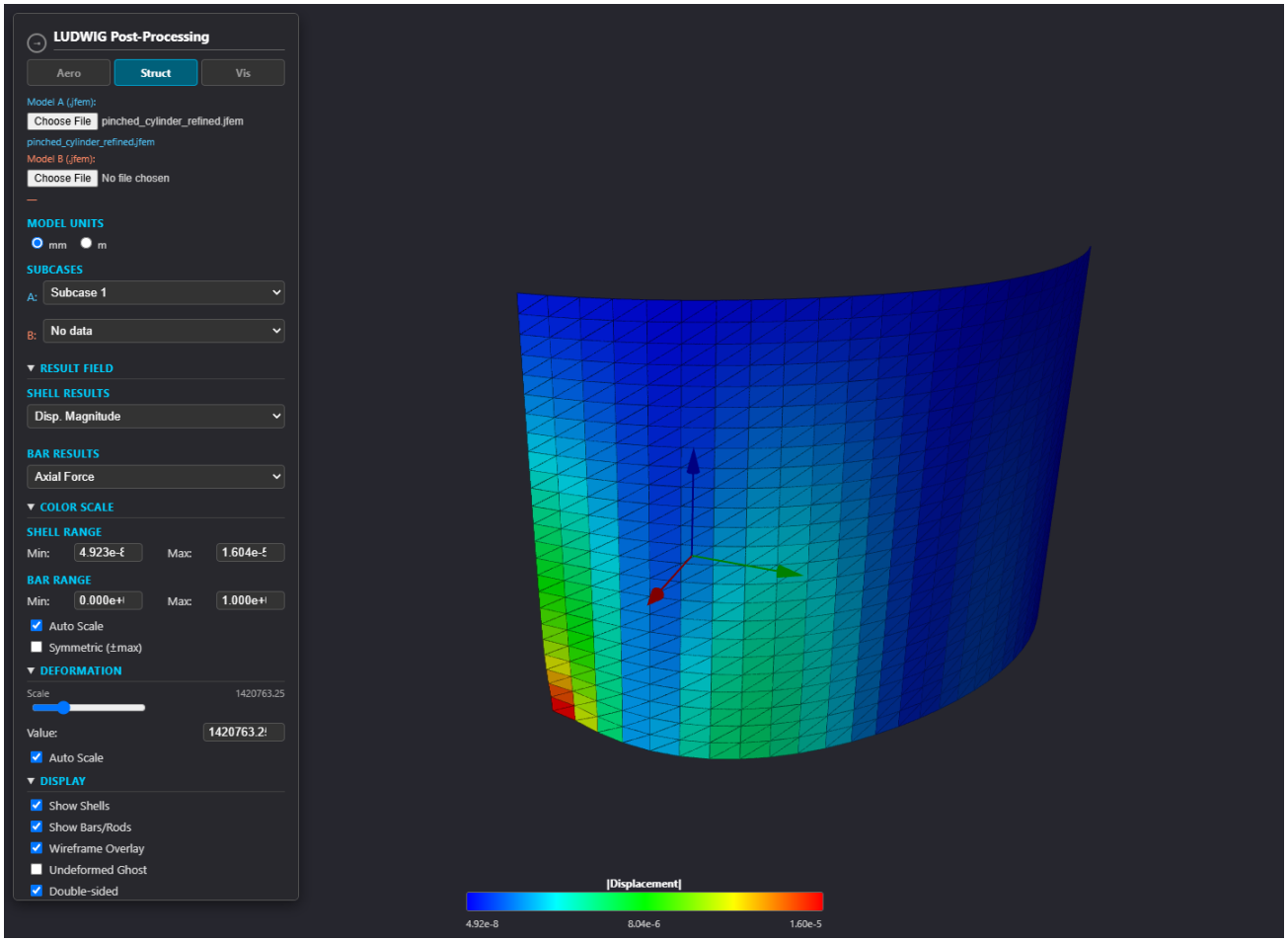
In JFEM-Boxer create the mesh, then use **Analysis / FE Analysis**. Define a solution and click **Run in JFEM**. Sensitivities are under **Analysis / Sensitivity**.

To try standalone JFEM, double-click **JFEM-solver.cmd** and select:

`__Starthere!__/examples/precompile/sol101_quad_static.bdf`

The console reports the output location. Open **Results-viewer.cmd**, select **Struct**, then **Choose File** for Model A and load the resulting `.jfem`.

Choose the case, result field and deformation scale. The viewer reads results; it does not solve models.



Worked example: the public refined pinched-cylinder model (576 shells), displacement magnitude, mesh edges and an automatically amplified deformation. The field legend shows the actual values; amplification is only visual.

6 Automatic fast startup

One workspace, one dependency environment and one compiled image for the solver and apps.

1. **JFEM-Boxer, standalone JFEM, PANDEATOR and analysis/sensitivity workers** load `sysimage/OpenJFEM_sysimage.dll`.

2. The root Project/Manifest pins the same packages for every app. No separate application environment or solver checkout is needed.

3. Launchers check source, dependencies and Julia version before loading an image. The console reports the selected image.

4. After updating the checkout or Julia, rerun **setup.cmd**. Do not force an old image with `-J`.

The image avoids repeating compilation covered by the installation workloads. New models or code paths can still require first-use compilation. Linux images use `.so`; macOS uses `.dylib`.

macOS / Linux

```
bash setup.sh
bash JFEM-Boxer.sh
bash PANDEATOR.sh
bash Results-viewer.sh
bash JFEM-solver.sh
```

On macOS the matching `.command` files are also provided. Julia must be on PATH. Keep the complete folder layout.

Advanced options

```
setup.cmd --dry-run
setup.cmd --sysimage-threads=4
setup.cmd --no-sysimage
```

The last option deliberately skips image construction. It is not needed for an ordinary installation.

More detail: `__Starthere!__/INSTALLATION.md` and `__Starthere!__/ADVANCED_SETUP.md`. The illustrated JFEM-Boxer guide is under `Reference_documentation/00_Getting_Started/`.